

Synthetic Market Phase 10

Context-conditioned geometry deformation. This phase keeps the same 4-asset latent market and asks whether stronger settings/context preserve the base coordinate system while deforming later set geometry in a structured way.

DATE	PROMPTS	CONTEXTS	OBJECT
22 March 2026	288 total	market-only, high-risk	low-risk, 4-asset geometry under settings deformation

MAIN READ

Phase 10 supports a preserve-then-deform story: the model keeps a common 4-asset market coordinate frame across settings, then later states bend that geometry toward score-like structure rather than replacing it.

The strongest clean positive is cross-context transfer: market-only probes still recover the latent axes in both risk-conditioned prompts with R^2 around 0.995 to 0.997. By row_eos around layers 12 to 13, the internal pair-distance geometry becomes slightly closer to the context-adjusted score space than to the raw latent layout. The cleanest warp is market-only to low-risk; the high-risk shift is real, but less directionally stable.

BEST X TRANSFER

0.99469

market -> low, row_mean L2

BEST Y TRANSFER

0.99678

market -> low, row_mean L2

BEST REALIGNMENT

+0.02323

low-risk, row_eos L12

CLEANEST WARP

0.27738

market -> low, cosine 0.699

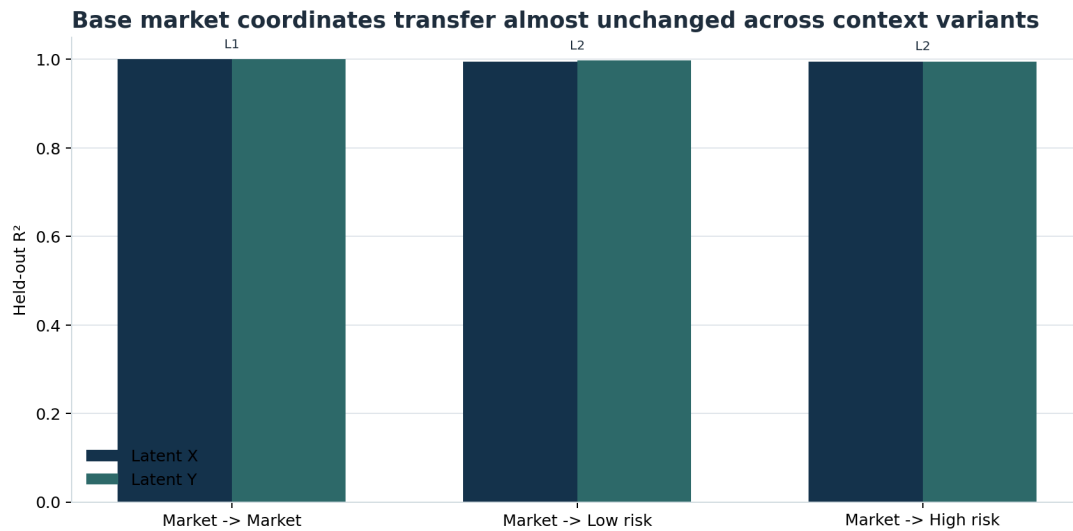
Why Phase 10 Exists

Phase 9 established a better object than row retrieval: the model clearly preserves per-asset coordinates and some within-snapshot geometry. The next question is whether later context:

- leaves that geometry alone
- destroys it
- or deforms it in a structured way

Phase 10 keeps the same 4-asset latent markets and adds stronger settings variants so we can separate *preservation* from *deformation*.

Coordinate Transfer Across Contexts



Market-only probes test whether the same latent axes remain recoverable after low-risk and high-risk settings are added.

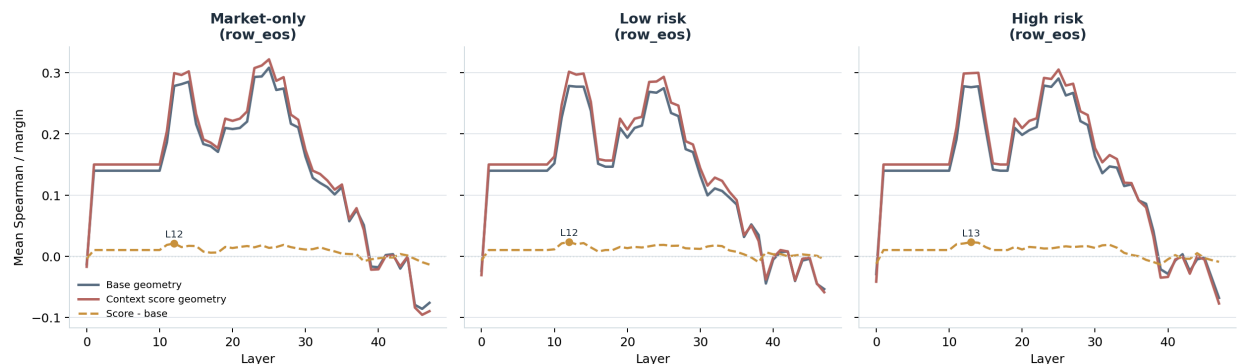
The transfer result is the most important anchor in the report. A probe trained only on market-only rows still recovers the latent coordinate system in the risk-conditioned prompts with very little degradation:

- latent_x: 0.99469 on market -> low-risk and 0.99422 on market -> high-risk
- latent_y: 0.99678 on market -> low-risk and 0.99491 on market -> high-risk

All of those peaks occur in early row_mean states. So the settings variants do not erase or replace the base market frame. The same underlying 4-asset coordinate system is still present and linearly recoverable after context is added.

Context Can Realign Later Geometry

Later states can realign toward context-adjusted geometry without losing the base coordinate system



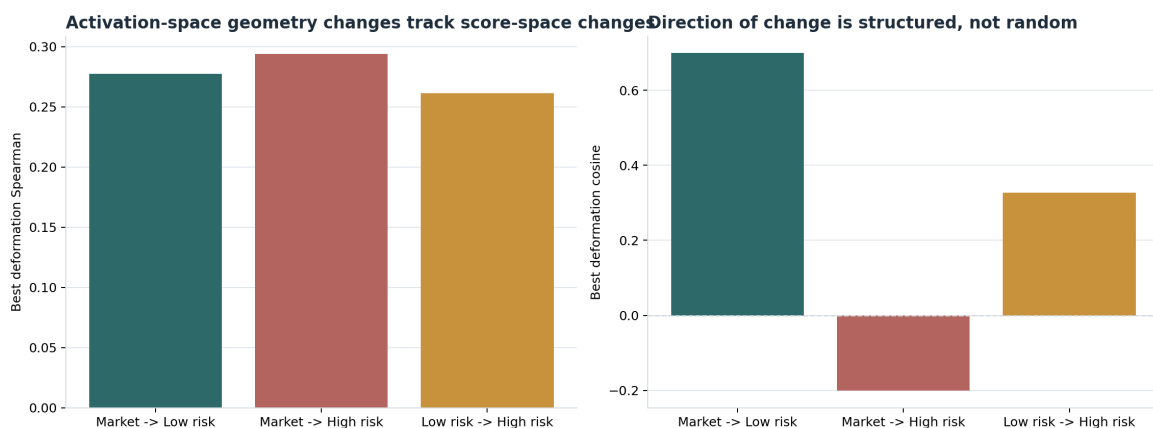
For each context, later states are compared against both the raw latent geometry and the context-adjusted score geometry.

The realignment effect is smaller, but it is the real Phase 10 gain over Phase 9. By row_eos around layers 12 to 13, the activation-space pair distances become modestly closer to the context-adjusted score geometry than to the raw latent layout in all three contexts:

- market_only: +0.02088
- low_risk: +0.02323
- high_risk: +0.02296

At the low-risk peak, the score-space distance Spearman is 0.30179 versus 0.27855 for the base latent geometry. That means later states are not just preserving the original hand-designed coordinates. They are compressing or rotating the market into a more score-like geometry. The fact that market_only shows the same qualitative shift matters: settings do not invent this transformation from nothing, they operate on top of a broader late-stage move toward score-space structure.

Geometry Deformation Is Structured



Matched prompt pairs isolate how the same market moves under changed settings, then compare activation-space movement to score-space movement.

The deformation metrics show whether matched prompts move through activation space in the same direction that their score-space geometry moves.

The cleanest case is market_only -> low_risk:

- deformation Spearman 0.27738
- deformation cosine 0.69926
- best state row_mean @ L2

That is a coherent early warp: the same market shifts in nearly the same direction that the synthetic score geometry predicts.

The market_only -> high_risk case is more mixed:

- deformation Spearman 0.29405
- deformation cosine -0.20079
- best state row_eos @ L1

So the ordering of pair-distance changes still tracks the score-space change, but the vector direction itself is not cleanly aligned.

Finally, low_risk -> high_risk peaks later:

- deformation Spearman 0.26131
- deformation cosine 0.32667
- best state row_eos @ L38

That suggests stronger setting deltas are handled further downstream than the softer market-to-low-risk adjustment.

Interpretation

SUPPORTED NOW

The model retains a common market coordinate frame across settings, and later context deforms that geometry in a measurable, structured way rather than wiping it out.

NOT SUPPORTED

A full geometry rewrite is not supported. Nor is a uniformly clean deformation story across all context pairs: the high-risk branch still looks partly mixed, with positive rank-like deformation but negative cosine at its best layer.

The best synthesis is:

- early row states preserve the base market axes
- later row_eos states compress those axes toward score geometry
- settings add structured warps on top of that compressed market space

That is a much stronger representational result than another pairwise or family-retrieval win. It suggests the model has a reusable multi-asset market frame, and context acts more like a deformation operator than a replacement representation.

What To Do Next

The next phase should stay set-level and make the deformation hypothesis more geometric and less probe-summary-driven:

1. Replace the discrete market_only / low_risk / high_risk split with a graded risk ladder and test whether the geometry rotates smoothly.
2. Add portfolio and affordance overlays to the same 4-asset base market and measure whether those produce new deformations or mostly gate the existing geometry.
3. Move from pair-distance summaries to explicit alignment transforms, for example Procrustes maps or learned linear maps between context-specific geometries.